CHIMA CYRIL HAMPO

Philadelphia, PA | +1 (215) 989-7648 | cch322@drexel.edu | linkedin.com/in/chimahampo | Google Scholar | github.com/hampochimacyril

RESEARCH INTERESTS

Indoor heat and humidity exposure in residential buildings; building-performance simulation (EnergyPlus/ResStock-style stock modeling); heat-health equity and health-linkage-ready exposure metrics; adaptive thermal comfort and overheating; hybrid physics-based / data-driven modeling of indoor climate; representative-location selection for national simulation workflows; life-cycle assessment of energy systems.

EDUCATION

Ph.D., Architectural Engineering — Drexel University, Philadelphia, PA — *Expected 2026*
Minor: Business Analytics. Focus: building-energy simulation, indoor thermal/moisture conditions, heat-stress mitigation, adaptive thermal comfort, resilience analytics. Funded by NIH Grant 1R01HL164726-01. Advisor: Prof. Simi Hoque; public-health collaborator: Dr. Leah H. Schinasi.

M.Sc., Mechanical Engineering (by Research) — Universiti Teknologi Petronas, Malaysia — *2019–2021*
LCA and thermodynamic modeling of district cooling and chiller systems; thermal energy storage.

B.Eng., Mechanical Engineering — University of Benin, Nigeria — *2013–2017*

Diploma, Mechanical Engineering Technology — Petroleum Training Institute, Nigeria — *2011–2013*

HONORS, AWARDS & GRANTS

- PDC Student Design Challenge — **1st Runner-Up (national), 2025**
- HackSimBuild (IBPSA-USA hackathon) — **Most Popular Hack** & **Most Creative Demonstration** (2026)
- NIH Graduate Research Grant, 1R01HL164726-01 (2022–2027)
- Research featured in **The New York Times** (2024) — indoor overheating and climate resilience
- Yayasan UTP Education Grant (M.Sc. Research Scholarship); UTP Graduate Assistantship Scheme (2019–2021)

PEER-REVIEWED PUBLICATIONS (SELECTED; † FIRST AUTHOR — FULL LIST, 20+, ON GOOGLE SCHOLAR)

1. † Hampo, C. C., Schinasi, L. H., & Hoque, S. (2025). Indoor temperature and humidity prediction in residential buildings: a review of white-box and black-box modeling techniques. *Building and Environment*, 114125.
2. † Hampo, C. C., Schinasi, L. H., & Hoque, S. (2024). Surviving indoor heat stress in the United States. *Heliyon*, 10(3).
3. Edwards, J. R., De Roos, A. J., Hampo, C. C., et al. (2025). Residential indoor temperatures and health: a scoping review of observational studies. *Science of the Total Environment*,

979, 179377.

4. Edwards, J. R., De Roos, A. J., Hampo, C. C., et al. (2024). Residential indoor temperatures and health: a systematic review of observational studies. *Environmental Health Perspectives Supplements*, 2024(1).
5. † Hampo, C. C., Akmar, A. B., & Majid, M. A. A. (2021). Life cycle assessment of an electric chiller integrated with a large district cooling plant. *Sustainability*, 13(1), 389.
6. † Hampo, C. C., Ya, H. H., Majid, M. A. A., Mokhtar, A. A., Rasangika, A. H. D. K., et al. (2021). Life cycle assessment of a vapor compression cooling system integrated within a district cooling plant. *Sustainability*, 13(21), 11940.
7. † Hampo, C. C., Mokhtar, A. A., Muhammed, M., Rasangika, K. A. H. D., & Shanshan, Y. (2022). Life cycle cost analysis of an electric centrifugal chiller integrated with a district cooling plant. *International Journal of Sustainable Energy*, 41(8), 1021–1036.
8. † Hampo, C. C., Ojo, D. O., Olatunde, D. E., et al. (2022). Life cycle assessment of renewable energy technologies in Northern Africa: a critical review. *Energy Sources, Part A*.
9. Abd Majid, M. A., Muhammad, M., Hampo, C. C., & Akmar, A. B. (2020). Analysis of a thermal energy storage tank in a large district cooling system. *Processes*, 8(9), 1158.
10. Oni, O. O., Nevo, C. M., Hampo, C. C., et al. (2022). Industrial symbiosis in Africa: status, challenges, prospects. *International Journal of Environmental Research*, 16(4), 49.
11. Muhammed, M., Mustapha, M., Ginta, T. L., Ali, A. M., Mustapha, F., & Hampo, C. C. (2020). Statistical review of microstructure–property correlation of stainless steel. *Processes*, 8(7), 811.
12. Joju, J., Verdi, D., Han, W. S., et al., Hampo, C. C., et al. (2023). Sustainability assessment of feedstock powder reuse for directed laser deposition. *Journal of Cleaner Production*, 388, 136005.
13. Echendu, A. J., Hampo, C. C., Olatunde, D., et al. (2024). Biofuels as a key renewable energy source: a review of LCA studies in South Africa. *Biofuels*, 15(5), 605–613.
14. Elijah, E. U., Ojo, D. O., Yakubu, E. A., Ezeuzo, E. B., Hampo, C. C., & Abubakar, A. (2023). LCA of solid waste management systems in African countries. *Journal of Advances in Environmental Health Research*, 11(4), 194–204.
15. Olatunde, D. E., Hampo, C. C., Akinlawon, A. P., Odobulu, O. B., & Afanwoubo, B. J. (2025). Africa's decarbonization strategies: pathway to net-zero by 2050–2060. *Discover Environment*, 3(1), 103.

Additional peer-reviewed articles listed on Google Scholar.

CONFERENCE PRESENTATIONS

- † Hampo, C. C., Schinasi, L. H., & Hoque, S. (2026). Overheating in Residential Buildings in the United States. SimBuild 2026 (IBPSA-USA Building Simulation Conference), Minneapolis, MN.
- † Hampo, C. C. (2025). Hybrid Modelling of Indoor Climate in Homes. Oral presentation, ISES-ISEE 2025 Joint Annual Meeting, Atlanta, GA.
- † Hampo, C. C. (2024). Quantifying Indoor Thermal Exposure: A Physics-based Approach. Pecha Kucha, 36th ISEE Annual Conference, Santiago, Chile.
- Edwards, J. R., De Roos, A. J., Hampo, C. C., et al. (2024). Residential Indoor Temperatures and Health: A Systematic Review. Poster (OP-093), ISEE 2024, Santiago, Chile.

SOFTWARE & RESEARCH TOOLS

ReplocX — full-stack decision-support app for representative U.S. location selection (Python, FastAPI, React/TypeScript, Docker, CI/CD; replocx.onrender.com; github.com/hampochimacyril/Location-representation-explorer). **AquaFlow** — award-winning whole-building water management platform (HackSimBuild / IBPSA-USA), where I led the AI and React 3D front-end. Modeling/analysis: EnergyPlus, OpenStudio, DesignBuilder, Revit, Insight; Python, R, MATLAB, SQL; openLCA, Umberto LCA+, ecoinvent.

LEADERSHIP & SERVICE

President (2025–26), Vice President (2024–25), Event Coordinator (2023–24), Graduate Student Association, Drexel University. Member: ASHRAE; ASHE. Teaching: lectures, tutorials, and laboratory sessions in engineering and building-systems topics; peer mentoring on simulation and reproducible reporting.

REFERENCES

Simi Hoque, PhD, PE, LEED AP — Professor & Program Director, CAEE, Drexel University · sth55@drexel.edu (PhD advisor) | **Leah H. Schinasi, MSPH, PhD** — Assistant Professor, Dornsife School of Public Health, Drexel University · lhs36@drexel.edu | **Michael S. Waring, PhD** — Professor & Department Head, CAEE, Drexel University · msw59@drexel.edu